Hedging or Harvesting?

Classical versus Learned Option Hedging under Transaction Costs

Replications of Sepp (2013) and Buehler et al. (2019), with a head-to-head comparison

DSC4107 Final Project

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Main body under 2000 words. Source code in Appendix B.
All results generated by the scripts in `submission/code/`.

Abstract

We reproduce two attacks on the same problem: hedging a short call when rebalancing is discrete and each trade costs money. Sepp (2013) derives closed-form profit-and-loss statistics for delta hedging and finds the frequency and bands that maximise the Sharpe ratio. Buehler et al. (2019) replace that family with a neural network trained on a convex risk measure. We rebuild both on one simulator, four market models and a shared cost schedule, then compare them. At zero cost the network recovers the Black-Scholes delta and the two approaches tie. At the 20 bp benchmark a well-tuned classical band matches or slightly beats the network. Larger Sharpe advantages above 50 bp arrive with a P&L standard deviation that triples, and a participation-dial extension shows the network is then a slightly worse premium harvester than not hedging at all. The one regime where a small edge survives every robustness check is Heston at 20 bp, and there it buys a 24% worse conditional tail.

1. Background and introduction

A desk sells an at-the-money one-month call and hedges it with the underlying. Trades happen at discrete times and each one is charged a proportional cost. The hedge therefore involves a trade-off: frequent rebalancing tracks the delta but pays more cost, and infrequent rebalancing saves cost but leaves variance.

Sepp (2013) studies that trade-off analytically. For four asset dynamics he derives closed forms for the mean P&L, the expected cost and the P&L variance of a time-scheduled delta hedge. He then maximises the Sharpe ratio, which has an interior optimum once cost is positive, and converts the optimum into equivalent price and delta bands. Buehler et al. (2019) take the opposite route. They keep the same frictions and let a neural network choose the hedge from market state, training it to minimise a convex risk measure of terminal P&L. The learned policy is not restricted to a fixed schedule or band, so it can trade less or hedge only partially.

The two papers are usually presented as opposite ends of a spectrum, with the network expected to dominate because it has more freedom. That expectation has not been measured under controlled conditions: same paths, contract and costs, with a classical baseline tuned separately for each regime. This project does that measurement. It also asks whether "the network wins" is the right summary, since a strategy can raise Sharpe by refusing to hedge and collecting the option premium instead.

2. Research questions

RQ1. Can a Monte Carlo simulator reproduce Sepp's closed-form statistics, tables and figures?

RQ2. Can we reproduce the main claim of Deep Hedging, that a trained network beats Black-Scholes delta hedging under transaction costs in a Heston market?

RQ3. On one testbed, which approach gives the better risk-adjusted P&L, and does the ranking change with the market model and the size of the transaction cost?

3. Methodology

Markets and contract. We reuse Sepp's four return processes: geometric Brownian motion (GBM), a Merton jump-diffusion (JD), Heston stochastic volatility (SV) and Heston with jumps (SVJ). Parameters are Sepp's own, so the replication lines up with his tables. Every experiment sells one at-the-money one-month call and hedges with the underlying. A proportional cost c is charged on the hedge trade, including the opening and closing trades, so a strategy cannot hide a trade from the accounting.

Sepp replication. `hedging/sepp.py` implements the closed forms for the mean coefficient u, the cost coefficient c, the local-variance term p and the variance floor f, then maximises $S(N) = \left(u - c\sqrt{N}\right) / \left(\sqrt{T}\sqrt{p/N + f}\right)$ over the number of rebalances N. The optimum maps to a price band and a delta band. We check the coefficients against Sepp's Table 2 and the statistics against a 60,000-

path Monte Carlo simulation re-run at the analytic optimum. Where they disagree, the formula is the candidate to debug; that is how the jump-diffusion factor in Section 4.1 was found.

Learned hedger. `hedging/deep.py` is a PyTorch multilayer perceptron. The state at each step is log-moneyness, time to maturity and current position; the network outputs the position to hold next. Training minimises the conditional value at risk (CVaR) of terminal P&L at tail fraction $\alpha = 0.65$. The learned hedger uses the same rollout as the classical strategies, so cost accounting is identical on both sides.

Benchmark and comparison. The frozen benchmark is Heston at a 20 bp proportional cost. The option is sold at 25% implied volatility against a 20% long-run realised level. Without that volatility risk premium, refusing to hedge dominates and a Sharpe comparison becomes meaningless, a failure mode we observed and removed. The classical baseline is a delta band (`delta_band = 0.1`) tuned once on a dedicated seed pool. For the head-to-head map we cover four market models and five cost levels. The classical side is re-tuned in every cell so a stale setting cannot manufacture a network win. The network trains and is scored on disjoint seed pools, and an Optuna add-on repeats four cells on a fourth, search-only pool.

Overfitting controls. Five seed pools (train, validation, tuning, search and holdout) are disjoint. Every reported number averages three independent training seeds, and we publish the per-seed minimum and win rate beside the mean. The deep and classical P&L standard deviations are printed together, so a Sharpe gain from under-hedging is visible. The holdout pool was simulated once, after every choice was frozen. The correctness anchor is the frictionless check: at zero cost the learned policy must recover the Black-Scholes delta.

4. Results and discussion

4.1 Sepp replication

The closed forms track Table 2. The largest relative error in u is 3.9%, and the GBM case I pair (0.5693 against 0.5692) and the SVJ case I pair (0.6551 against 0.6550) differ by 0.0001. The optimum is interior and moves to lower frequency as the model gains complexity: extra trading cannot remove a variance floor f, so paying more cost buys less. GBM case I optimises at 242 rebalances with a Sharpe of 0.83; Heston case I at 75 with 0.43.

The jump-diffusion row needed a real fix. The JD branch had dropped a factor $q = \sqrt{\pi^3}/4$ that the GBM and SVJ branches carry. `scripts/diagnose_jd.py` shows the second-moment gamma was correct to 0.2% all along; only the coefficient was wrong. Restoring q moves p from 24% below the paper to 5.6% above it and brings the analytic JD Sharpe (0.65) onto the simulated one (0.65) in case I. The remaining case-II gap of about 12% in volatility is Sepp's own variance approximation. The Monte Carlo cross-check passes for the other three models in both mean and volatility.

4.2 Deep hedging replication

At zero cost the learned policy rediscovers the delta. Over 20,000 paths the correlation between learned position and Black-Scholes delta is 0.9907, with slope 1.148 and RMSE 0.0581; the mean learned position (0.499) is within 2% of the mean delta (0.508).

The Heston experiment then reproduces the paper's qualitative claim. At zero cost the network matches a delta hedge (Sharpe 1.154 against 1.139). As cost rises it moves ahead of both every-step delta hedging and the frozen band.

Cost (bp)	Learned	BS delta	Frozen band	Edge vs delta	Edge vs band
0	1.154	1.139	0.987	+0.016	+0.167
20	0.373	0.254	0.352	+0.119	+0.020
50	0.105	−0.970	−0.526	+1.075	+0.631

Cost (bp)	Learned	BS delta	Frozen band	Edge vs delta	Edge vs band
100	0.182	−2.276	−1.573	+2.458	+1.755

Table 1: Heston Sharpe by cost. The learned hedger pulls ahead as cost rises; Section 4.3 shows what its P&L spread does at the same time.

4.3 The head-to-head comparison

The edge map has three regimes, and the P&L standard deviations explain the third.

Near zero cost the two approaches tie, which is the expected outcome once the frictionless check passes. At 20 bp, the benchmark's own cost, the comparison is flat: edges run from −0.09 to +0.02, and on the jump models the network is behind. Above 50 bp the network's Sharpe edge becomes large and consistent, but its P&L standard deviation climbs from about 0.5 to 3 or 4 while its mean P&L approaches the option premium. The network has learned to stop hedging and collect the volatility risk premium. That is a valid strategy, though a different product from a hedge; a Sharpe win produced by not hedging is a change of risk profile.

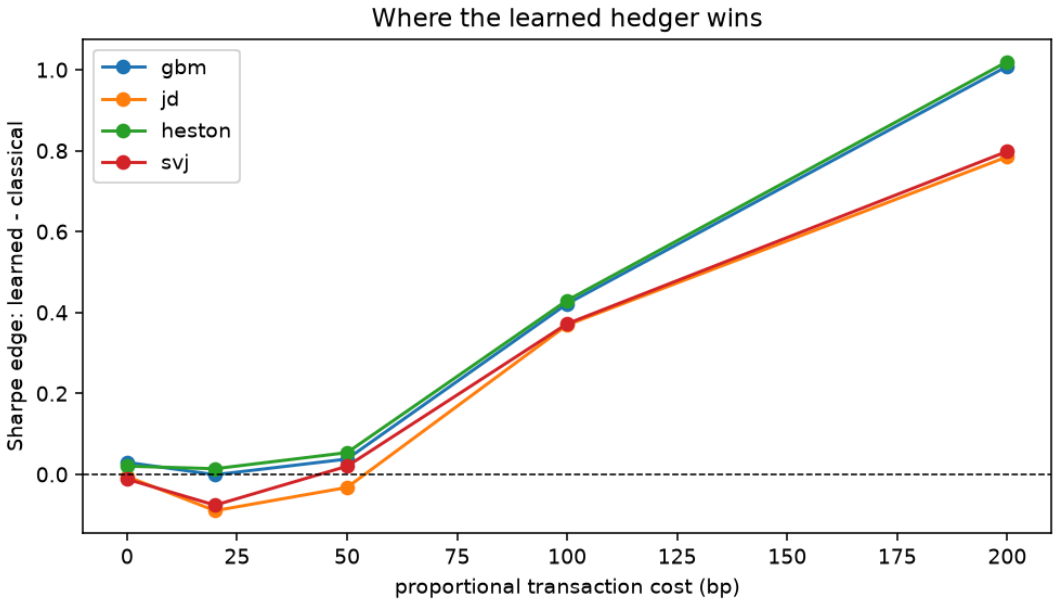


Figure 1: Learned minus classical Sharpe. High-cost wins arrive with a tripled P&L spread.

The locked holdout confirms the moderate-cost result and its size. On five unseen seeds the mean edge over the frozen band is +0.0339, with a minimum of +0.0293 and a win rate of 1.00. Bootstrap intervals on the validation pool exclude zero, so the small Heston 20 bp edge is not seed noise or path noise.

It is also not a free improvement. On that cell the two P&L distributions have almost the same spread, but the learned one has skew −0.70 against −0.01 and a CVaR at 95% of 1.321 against 1.066, which is a 24% worse tail. Whether the edge counts as a win depends on whether the desk scores Sharpe or tail risk, and the two measures disagree.

4.4 What the follow-up probes add

Three probes sharpen the mechanism. Sweeping the CVaR tail fraction α shows it is the dial between hedging and harvesting: as α goes to 1, turnover collapses from about 180 to about 2 and the P&L standard deviation triples. Removing the current position from the state costs 0.056 Sharpe and raises turnover, so position feedback is what produces band-like behaviour. The learned policy also transfers across markets: training on GBM loses only about 0.02 Sharpe on Heston, which makes the ranking across markets a property of the market rather than of overfitting to training paths.

A sidequest tests whether the high-cost win is only a position-size choice. We add a participation dial λ to the classical family, holding λ times the delta and rebalancing every k steps, which contains the textbook delta ($\lambda = 1$) and the do-nothing hedge ($\lambda = 0$). At 0 to 20 bp the tuned dial sits near $\lambda = 1$ at a slower calendar and the network stays 0.02 to 0.05 above it. Above 20 bp the dial collapses to $\lambda = 0$, and there the network sits below do-nothing in every high-cost cell. Its high-cost Sharpe edge over the report's classical family is therefore an artefact of that family being forced to open a full delta at $t = 0$.

4.5 Limitations

This is a course-scale testbed: 80 to 100 epochs, 2,048 training paths per seed, three seeds, a small network and eight Optuna trials per cell. The direction of the results is stable across fixed and searched configurations, but the exact edges are not paper-grade magnitudes. The classical family in the map always opens a full initial hedge, so it cannot express "under-hedge" or "hold flat"; the sidequest repairs that for interpretation but stayed outside the frozen benchmark. The comparison scripts are covered by smoke tests rather than the unit suite, so their results rest on seed discipline.

5. Conclusion

The extra freedom of a neural hedger does not buy a better hedge in the benchmark regime. At 20 bp the trained network ties a well-tuned classical delta band, and the small Heston edge that survives the holdout comes with a worse tail. Where the network does post large Sharpe gains, above roughly 50 bp, it has stopped hedging and is collecting the volatility risk premium, and a classical family given the same freedom does that better. The two approaches differ less in hedging quality than the headline table suggests, and any claim that a learned hedger wins should be reported next to its P&L spread. The replications themselves hold up: the Sepp coefficients match Table 2, the closed forms agree with Monte Carlo, and the frictionless network recovers the Black-Scholes delta.

References

[1] H. Buehler, L. Gonon, J. Teichmann and B. Wood. Deep hedging. *Quantitative Finance*, 19(8):1271-1291, 2019. doi:10.1080/14697688.2019.1571683.

[2] A. Sepp. When you hedge discretely: Optimization of Sharpe ratio for delta-hedging strategy under discrete hedging and transaction costs. *Journal of Investment Strategies*, 3(1):19-59, 2013.

Appendix A. Extended tables

A.1 Sepp Table 2 reproduction

Case	Model	u ours	u paper	p ours	p paper	N* ours	N* paper	S ours	S paper
I	GBM	0.5693	0.5692	0.2519	0.2461	242	239	0.83	0.89
I	JD	0.5640	0.5510	0.2118	0.2006	147	138	0.65	0.66
I	Heston	0.6621	0.6703	0.2583	0.2524	75	74	0.43	0.45
I	SVJ	0.6551	0.6550	0.2164	0.2085	70	68	0.43	0.45
II	GBM	1.8080	1.8558	0.6138	0.5998	130	129	1.18	1.30
II	JD	1.7826	1.7628	0.4765	0.4423	88	75	1.01	0.99
II	Heston	2.0054	2.0876	0.6359	0.6214	49	49	0.65	0.73
II	SVJ	1.9738	1.9995	0.4903	0.4634	45	43	0.68	0.72

Table 2: Closed-form coefficients against Sepp's Table 2. Largest relative error in u is 3.9%.

A.2 Monte Carlo cross-check at the analytic optimum

Case	Model	P&L analytic	P&L MC	vol analytic	vol MC	Sharpe a/MC
I	GBM	0.00318	0.00309	0.00385	0.00365	0.83 / 0.85
I	JD	0.00373	0.00381	0.00571	0.00590	0.65 / 0.65
I	Heston	0.00511	0.00537	0.01191	0.01173	0.43 / 0.46
I	SVJ	0.00515	0.00542	0.01187	0.01168	0.43 / 0.46
II	GBM	0.01098	0.01102	0.00932	0.00916	1.18 / 1.20
II	JD	0.01220	0.01295	0.01213	0.01370	1.01 / 0.95
II	Heston	0.01526	0.01695	0.02338	0.02369	0.65 / 0.72
II	SVJ	0.01539	0.01709	0.02280	0.02380	0.68 / 0.72

Table 3: 60,000-path cross-check. Mean matches for every model; the residual case-II volatility gap is the paper's own approximation.

A.3 Full edge map, fixed configuration

Model	Cost	Edge	Edge min	Win	Deep Sharpe	Classical Sharpe	Deep std / Cls std
GBM	0 bp	+0.030	+0.014	1.00	1.258	1.228	0.46 / 0.47
GBM	20 bp	−0.001	−0.008	0.50	0.347	0.348	0.55 / 0.58
GBM	50 bp	+0.038	+0.032	1.00	0.073	0.035	1.72 / 1.74
GBM	100 bp	+0.422	+0.416	1.00	0.161	−0.261	3.44 / 1.74
GBM	200 bp	+1.009	+1.004	1.00	0.158	−0.851	3.48 / 1.74
JD	0 bp	−0.006	−0.009	0.00	0.116	0.121	1.98 / 1.89
JD	20 bp	−0.090	−0.096	0.00	−0.081	0.008	1.94 / 2.47
JD	50 bp	−0.032	−0.037	0.00	−0.149	−0.117	2.19 / 2.47
JD	100 bp	+0.369	+0.361	1.00	0.044	−0.325	4.21 / 2.47
JD	200 bp	+0.785	+0.770	1.00	0.042	−0.743	4.37 / 2.46
Heston	0 bp	+0.020	+0.007	1.00	1.159	1.139	0.51 / 0.52

Model	Cost	Edge	Edge min	Win	Deep Sharpe	Classical Sharpe	Deep std / Cls std
Heston	20 bp	+0.014	+0.006	1.00	0.366	0.352	0.61 / 0.61
Heston	50 bp	+0.054	+0.047	1.00	0.100	0.046	1.60 / 1.74
Heston	100 bp	+0.430	+0.425	1.00	0.179	−0.251	3.11 / 1.73
Heston	200 bp	+1.020	+1.015	1.00	0.174	−0.846	3.17 / 1.73
SVJ	0 bp	−0.012	−0.016	0.00	0.113	0.125	2.11 / 1.92
SVJ	20 bp	−0.076	−0.082	0.00	−0.062	0.014	2.05 / 2.45
SVJ	50 bp	+0.020	+0.016	1.00	−0.092	−0.112	2.36 / 2.45
SVJ	100 bp	+0.372	+0.368	1.00	0.050	−0.322	3.97 / 2.45
SVJ	200 bp	+0.798	+0.789	1.00	0.054	−0.744	4.10 / 2.44

Table 4: Learned minus classical Sharpe. The standard-deviation columns are the guard rail: where the edge is large, the network has stopped hedging.

A.4 Optuna hyperparameter search

Model	Cost	Search edge	Reported edge	Win	Deep Sharpe	Classical Sharpe	Deep std / Cls std
Heston	20 bp	+0.011	+0.021	1.00	0.374	0.352	0.60 / 0.61
Heston	50 bp	+0.140	+0.139	1.00	0.185	0.046	3.09 / 1.74
Heston	100 bp	+0.436	+0.435	1.00	0.185	−0.251	3.18 / 1.73
JD	20 bp	−0.075	−0.077	0.00	−0.069	0.008	1.96 / 2.47
JD	50 bp	+0.158	+0.156	1.00	0.039	−0.117	3.98 / 2.47
JD	100 bp	+0.379	+0.377	1.00	0.052	−0.325	4.33 / 2.47

Table 5: Optuna search ranked on a disjoint search pool. The search does not change the qualitative picture.

A.5 Terminal P&L distribution on the frozen benchmark

Strategy	Mean	Std	Skew	CVaR 95%	5% quantile
Learned	0.240	0.626	−0.699	1.321	−0.880
Band	0.216	0.613	−0.010	1.066	−0.752

Table 6: Same spread, worse left tail. The +0.03 Sharpe edge is partly bought with a 24% worse CVaR.

A.6 Sidequest participation frontier

Model	Cost	Tuned lambda	Freq	Participation Sharpe	Learned Sharpe	Learned - part.	No-hedge Sharpe
GBM	0 bp	1.00	1	+1.228	+1.258	+0.030	+0.165
GBM	20 bp	1.00	3	+0.308	+0.347	+0.039	+0.165
GBM	100 bp	0.00	1	+0.165	+0.161	−0.004	+0.165
GBM	200 bp	0.00	1	+0.165	+0.158	−0.008	+0.165
JD	20 bp	0.00	1	+0.053	−0.081	−0.135	+0.053
JD	100 bp	0.00	1	+0.053	+0.044	−0.009	+0.053

Model	Cost	Tuned lambda	Freq	Partic-ipation Sharpe	Learned Sharpe	Learned - part.	No-hedge Sharpe
Heston	0 bp	1.00	1	+1.139	+1.159	+0.020	+0.187
Heston	20 bp	1.00	3	+0.317	+0.366	+0.049	+0.187
Heston	100 bp	0.00	1	+0.187	+0.179	−0.007	+0.187
Heston	200 bp	0.00	1	+0.187	+0.174	−0.012	+0.187
SVJ	20 bp	0.00	1	+0.060	−0.062	−0.122	+0.060
SVJ	100 bp	0.00	1	+0.060	+0.050	−0.010	+0.060
SVJ	200 bp	0.00	1	+0.060	+0.054	−0.006	+0.060

Table 7: The extended classical family (participation dial lambda). Where the network posts a high-cost Sharpe win, the tuned dial is zero and the network is below do-nothing.

A.7 Figures

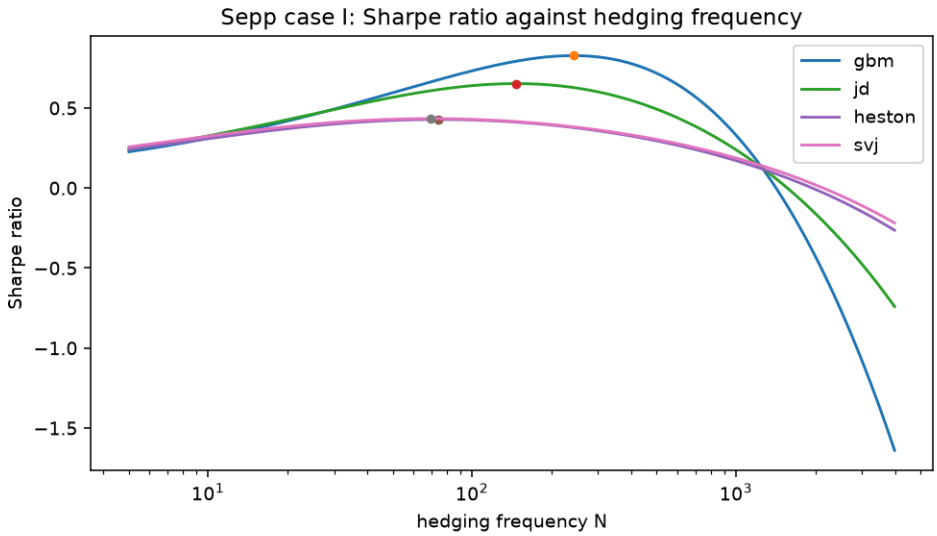


Figure 2: Sepp case I: Sharpe against hedging frequency for the four models. The optimum is interior once cost is positive.

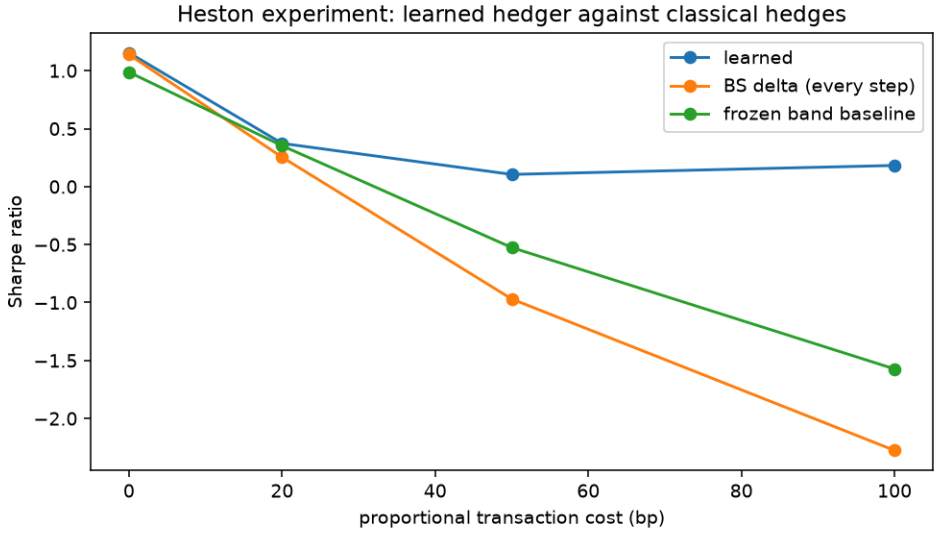


Figure 3: Heston deep-hedging replication: Sharpe against transaction cost for the learned, BS delta and band strategies.

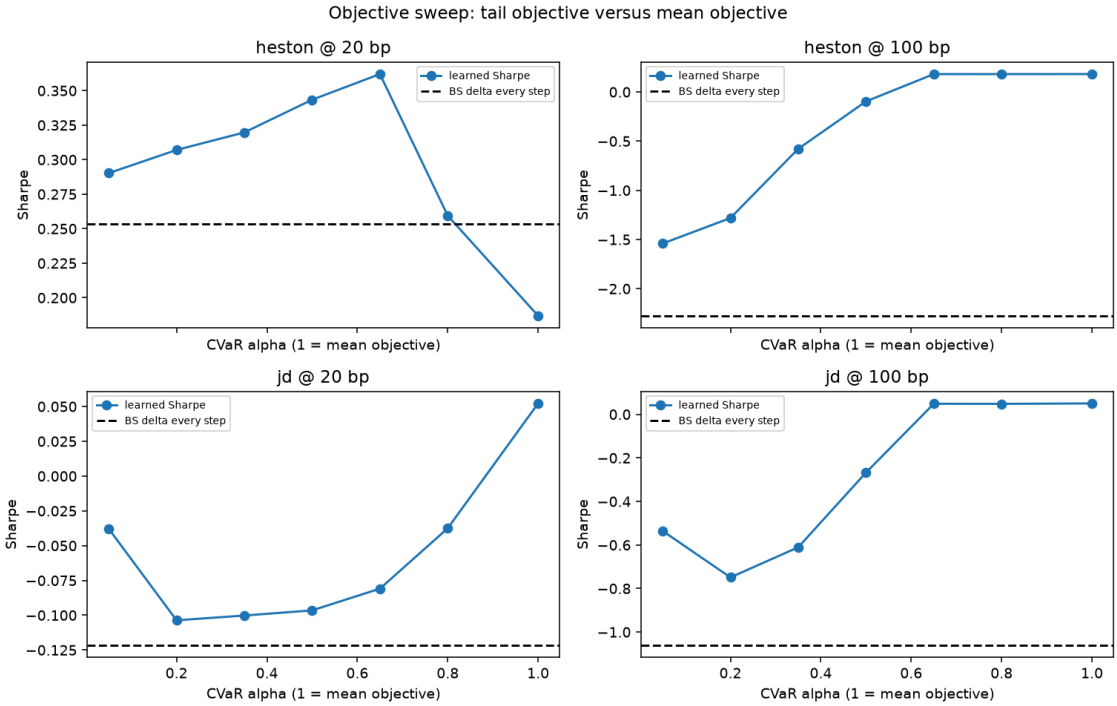


Figure 4: CVaR tail fraction as the hedging-versus-harvesting dial. As alpha tends to 1, turnover collapses and the P&L spread triples.

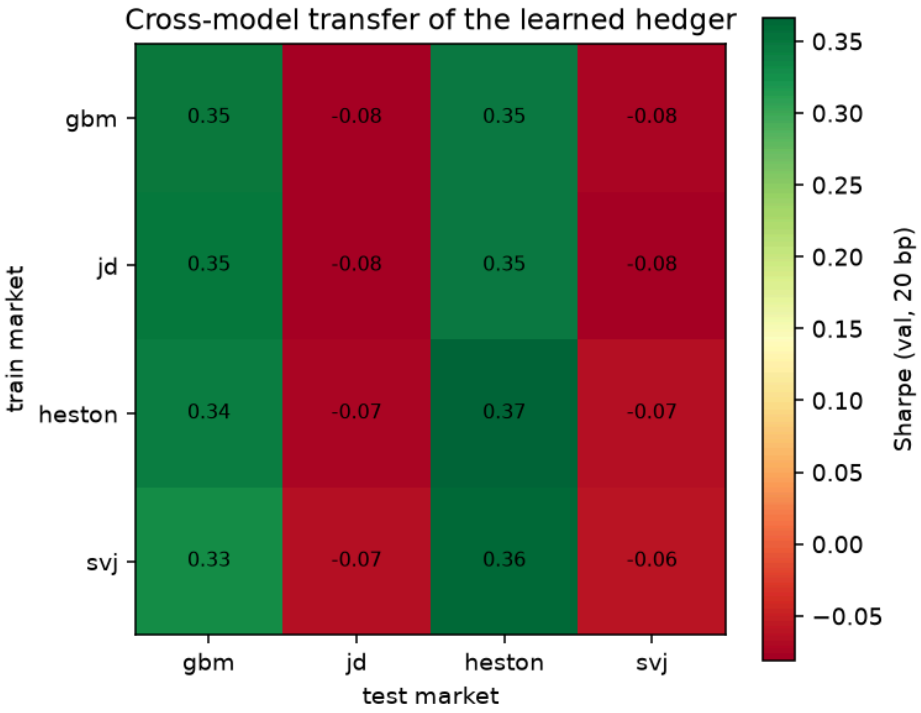


Figure 5: Cross-model transfer: training on one model and scoring on all four. The column order dominates, so the ranking is a property of the market.

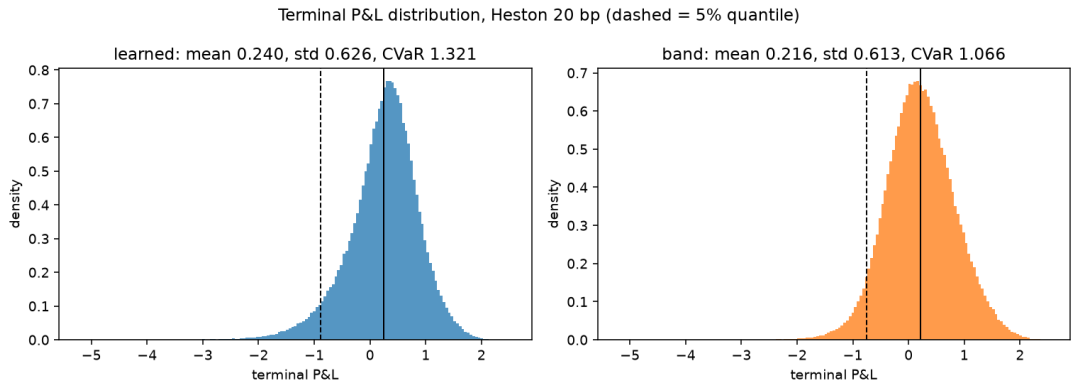


Figure 6: Terminal P&L on the frozen benchmark. The learned distribution has the same spread as the band but a heavier left tail.

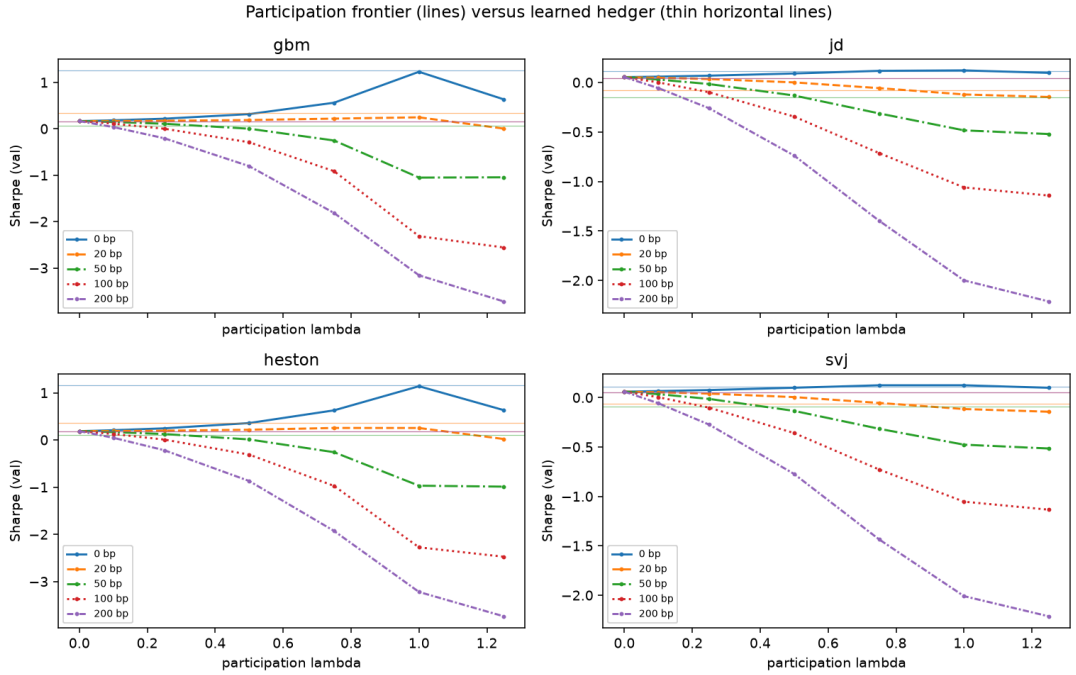


Figure 7: Participation frontier. Above 20 bp the tuned dial is zero and the network sits below do-nothing.

Appendix B. Source code repository

The complete source code is on GitHub:

`github.com/hluaguo/dsc4107`

The repository holds every module used to produce the numbers in this report: the four market simulators, the shared P&L rollout, the Sepp closed forms, the learned hedger and the comparison and exploration scripts. It also carries the test suite and the pinned environment. A snapshot is attached under `submission/code/`, and the raw experimental data sits under `submission/data/`.

Path	Role
hedging/market.py	GBM, JD, Heston and SVJ simulators (frozen)
hedging/bs.py	Black-Scholes price and greeks (frozen)
hedging/hedge.py	Shared P&L rollout and classical strategies (frozen)
hedging/metrics.py	Sharpe, CVaR and the reported statistics (frozen)
hedging/bench.py	Frozen benchmark config and seed pools
hedging/sepp.py	Sepp (2013) closed forms and Sharpe optimisation
hedging/deep.py	Learned hedger: features, network, CVaR loss
hedging/hyper.py	Training hyperparameters
scripts/sepp_replication.py	Sepp Table 2, optima and Monte Carlo check
scripts/deep_hedging_replication.py	Frictionless check and Heston experiment
scripts/comparison_study.py	Head-to-head map and Optuna add-on
scripts/holdout_eval.py	The single locked holdout run
`scripts/explore_*.py`	Objective, feature, transfer and robustness probes
scripts/sidequest_participation.py	The participation-dial test
tests/	Analytic and accounting checks (24 tests)

Table 8: Repository layout.

Appendix C. How to reproduce

The full environment is pinned in `uv.lock`. Clone the repository and sync it:

```
# from the repository root
git clone https://github.com/hluaguo/dsc4107.git
cd dsc4107
uv sync
uv run pytest -q                                          # 24 checks
uv run python scripts/sepp_replication.py                 # Sepp tables + figures
uv run python scripts/sepp_replication.py --mc --mc-paths 60000
uv run python scripts/deep_hedging_replication.py --epochs 100 --hidden 64 --depth 3
uv run python scripts/comparison_study.py --fixed-hyper \
    --costs 0.0 0.002 0.005 0.01 0.02
uv run python scripts/comparison_study.py --models heston jd \
    --costs 0.002 0.005 0.01 --trials 8
uv run python scripts/holdout_eval.py                     # final locked run
```

All training scripts select CUDA when it is present and fall back to the CPU otherwise. Raw experimental data and figures are mirrored under `submission/data/`.